

Spatial Structural Modeling

Empirical Methods — Week 17

Teaching deck draft

Week 17 question

- When do spatial research questions require equilibrium structure?
- Lecture 15 built spatial objects: places, distances, exposures, and access.
- Lecture 16 asked what identifies causal effects when space creates spillovers, sorting, or correlated shocks.
- Lecture 17 asks when workers, firms, rents, amenities, commuting, migration, and welfare must be modeled jointly.

Structure is useful when the policy object is an equilibrium outcome, not just a local treatment effect.

When reduced form is not enough

Reduced-form fact	Missing equilibrium object	Why it matters
Local wages rise	rents, amenities, commuting costs	nominal wages are not welfare
Employment changes in treated place	migration and commuting flows	local shocks do not stay local
Housing reform affects prices	housing supply and sorting	gains may accrue to workers or landlords
Transit lowers travel time	job access and firm response	residence and workplace incidence differ
Place subsidy raises jobs	firm location and spillovers	nearby places may lose or gain

Need structure when $\mathcal{E}(d) = \{w, r, F, L, H, U\}(d)$ is the object.

Core equilibrium map

$$V_{ioj} = w_j - r_i - c_{ij}^{commute} - m_{oi} + a_i + \varepsilon_{ioj}$$

$$P_{ioj} = \frac{\exp(\bar{V}_{ioj}/\sigma)}{\sum_{i'} \sum_{j'} \exp(\bar{V}_{oi'j'}/\sigma)}$$

$$\mathcal{E}(\theta; d) = \{P_{ioj}, \pi_{ij}^{commute}, w_j, r_i, L_j, H_i, U_o\}_{o,i,j}$$

- Workers choose locations and commutes given wages, rents, amenities, and frictions.
- Firms demand labor where productivity and market access are high.
- Housing supply and congestion determine rent incidence.
- Equilibrium closes through prices, quantities, and bilateral flows.

Core objects to keep separate

Object	Typical evidence	Main risk
Moving costs	migration flows, mover histories	confuse amenities with frictions
Commuting costs	residence-workplace flows, travel time	treat distance as generalized cost
Local productivity	wages, output, firm data	absorb amenities or sorting
Amenities	residual utility, housing demand	residual becomes everything
Rents and housing supply	prices, quantities, regulation, land	miss landlord incidence
Labor demand and firm location	employment, establishments, shocks	hold firms fixed when policy moves them

The model is credible only if these objects are not silently bundled together.

Static versus dynamic spatial structure

Question	Static model	Dynamic model
Main use	long-run incidence and comparative statics	transition paths and policy timing
Adjustment	workers, firms, rents, flows clear after shock	migration, housing, tenure, and investment adjust over time
Data pressure	cross-section, long differences, flows	panels, histories, timing, expectations
Best when	long-run allocation is the object	adjustment costs are the object
Main caveat	hides transition costs	heavier assumptions and computation

The static-dynamic choice should follow the estimand, not the desire for a larger model.

Gravity-style building blocks

$$F_{ij} = \frac{O_i D_j \phi_{ij}^{-\kappa}}{\sum_{\ell} D_{\ell} \phi_{i\ell}^{-\kappa}} \quad \log F_{ij} = \alpha_i + \delta_j - \kappa \log \phi_{ij} + e_{ij}$$

- Gravity disciplines bilateral trade, commuting, migration, applications, or trips.
- Observed flows help recover frictions and flow elasticities.
- The same algebra can sit inside many spatial models.
- Gravity is a building block, not the whole model, when the policy question is incidence or welfare.

Identification, calibration, and fit

Model object	Empirical discipline	Diagnostic
Commuting friction	bilateral residence-workplace flows	gravity residuals, untargeted flows
Migration cost	mover flows and histories	response to shocks, origin persistence
Productivity	wages, output, firm data	separate from amenities and composition
Housing supply	rents, quantities, regulation	sensitivity to elasticity
Amenities	residual utility or external measures	validate with non-targeted outcomes
Counterfactual	policy shock and fixed point	distance from observed support

Report what is estimated, what is calibrated, what is inverted, and what is held fixed.

Counterfactual interpretation

$$\Delta W = \sum_g \omega_g [U_g(\mathcal{E}(\theta; d_1)) - U_g(\mathcal{E}(\theta; d_0))]$$

- Welfare requires a utility object, not only wages or employment.
- Incidence can differ for residents, in-commuters, movers, firms, landlords, and taxpayers.
- Nearby policy changes are more credible than large reforms far outside support.
- Sensitivity should target the margins that move the headline result: frictions, housing supply, amenities, and firm response.

Spatial methods block summary

Lecture	Skill	Research payoff
15	curate spatial objects	places, distances, crosswalks, access, exposure
16	identify effects with space	spillovers, sorting, borders, shift-share, inference
17	model equilibrium propagation	wages, rents, flows, firms, housing, welfare

Where the literature is now:

- richer bilateral flow data and quantitative spatial models;
- more attention to equilibrium properties, multiplicity, and computation;
- dynamic urban models with transition paths, housing frictions, and forward-looking households;
- frontier questions on remote work, climate adaptation, monopsony in space, and heterogeneous workers.

Reproduce

- synthetic locations
- commuting gravity
- commuting openness
- local productivity shock

Diagnose

- targeted vs untargeted fit
- friction sensitivity
- housing supply sensitivity
- reduced-form stopping point

Transfer

- dynamic adjustment path
- migration friction
- gradual housing response
- static versus dynamic memo

Primary anchor: Monte–Redding–Rossi–Hansberg. Dynamic extension: Greaney–Parkhomenko–Van Nieuwerburgh.

- Spatial structural models are justified by the object, not by style.
- The core object is an equilibrium mapping from shocks to prices, flows, quantities, and welfare.
- Gravity-style blocks are valuable flow discipline inside that mapping.
- Identification, calibration, fit, and counterfactual distance should be reported separately.
- The spatial block now gives three tools: build spatial objects, identify spatial effects, and model equilibrium propagation.